Republic of the Philippines
ENERGY REGULATORY COMMISSION
Pasig City

RESOLUTION NO. 20, SERIES OF 2025

A RESOLUTION ADOPTING THE AMENDED RULES TO GOVERN THE IMPLEMENTATION OF ADVANCED METERING INFRASTRUCTURE (AMI) BY DISTRIBUTION UTILITIES AND ERC-AUTHORIZED ENTITIES

WHEREAS, Section 2 of Republic Act No. 9136, otherwise known as the Electric Power Industry Reform Act of 2001 (EPIRA), declares that it is the policy of the State to ensure and accelerate the total electrification of the country, to ensure the quality, reliability, security, and affordability of the supply of electric power, to protect the public interest as it is affected by the rates and services of electric utilities and other providers of electric power, and encourage the efficient use of energy and other modalities of demand side management;

WHEREAS, Section 16 (c) of Commonwealth Act No. 146, otherwise known as the Public Service Act, amended by Republic Act No. 11659,[1] mandates the Commission to fix just and reasonable standards, classifications, regulations, practices, measurement, or service to be imposed, observed, and followed thereafter by any public service;

WHEREAS, Section 16 (g) of Commonwealth Act No. 146, as amended, authorizes the Commission to compel any public service to furnish safe, adequate, and proper service as regards the manner of furnishing the same as well as the maintenance of the necessary material and equipment;

WHEREAS, Section 20 (b) of Commonwealth Act No. 146, as amended, requires approval from the Commission for distribution utilities to establish, construct, maintain, or operate new units or extend existing facilities or make any other addition to or general extension of the service;

WHEREAS, Section 41 of Republic Act No. 9136 mandates the Commission to ensure the adequate promotion of consumer interest;

WHEREAS, Resolution No. 01, Series of 2021, otherwise known as the Commission's Revised Rules of Practice and Procedure (RRPP), provides

[1] An Act Amending Commonwealth Act No. 146, otherwise known as the Public Service Act, As Amended.

for the rules to govern proceedings to be conducted for cases pending before the Commission;

WHEREAS, the Commission determined the necessity to streamline the process of regulatory approvals for the implementation of the Advanced Metering Infrastructure (AMI) project, to address regulatory gaps within the existing AMI rules, to harmonize the rules with new laws, jurisprudence, government policies, rules and regulations, and to update the existing AMI rules with technological advancements including risks thereto;

WHEREAS, the Commission issued an *Order* on 13 May 2022, directing all distribution utilities (DUs) to accomplish the AMI Survey Form. The survey was designed to assess the state of DUs' networks and their compliance with the Smart Grid Framework and Roadmap mandated by the Department of Energy (DOE) through Department Circular No. 2020-02-0003[1];

WHEREAS, the Commission engaged the assistance of USAID in securing technical consultants with expertise in AMI systems. These consultants conducted a review of the existing AMI Rules and submitted a draft revised AMI Rules on 15 December 2022;

WHEREAS, the Commission posted the proposed draft revised AMI Rules on the ERC Website on 08 May 2023 for purposes of soliciting comments from interested stakeholders and consumers, and for the conduct of public consultations;

WHEREAS, the Commission received comments from various stakeholders;

WHEREAS, the Commission conducted public consultations on 19 May 2023 (Luzon), 25 May 2023 (Visayas) and 30 June 2023 (Mindanao);

WHEREAS, in addition to the comments received and public consultations, the Commission consulted the Department of Information and Communication Technology (DICT), smart grid technology and cybersecurity practitioners and experts and reviewed the pertinent Circulars of the Department of Energy (DOE) on Smart Grid and Cybersecurity;

NOW, THEREFORE, the Commission, hereby **RESOLVES** to **APPROVE** and **ADOPT** the *Amended Rules to Govern the Implementation of Advanced Metering Infrastructure (AMI) by Distribution Utilities and ERC-Authorized Entities*, hereto attached as **Annex "A"** and made an integral part hereof.

This Resolution shall take effect fifteen (15) days following its publication in a newspaper of general circulation or in the Official Gazette.

Let copies of this *Resolution* be furnished to the University of the Philippines Law Center - Office of the National Administrative Register (UPLC-ONAR), as well as published on the Commission's official website and such other online platforms available to the ERC.

Pasig City.

FRANCIS SATURNINO C. JUAN
Chairperson and CEO

FLORESINDA G. BALDO-DIGAL
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MARKO ROMEO L. FUENTES
Commissioner

AMANTE A. LIBERATO
Commissioner

PARIS G. REAL
Commissioner

AMI-TWG: MCCG / MKCM / MCS / RFA / GDB

**Deliberated and approved during the 30 October 2025 Commission Meeting.*

Annex A

AMENDED RULES TO GOVERN THE IMPLEMENTATION OF ADVANCED METERING INFRASTRUCTURE (AMI) BY DISTRIBUTION UTILITIES AND ERC-AUTHORIZED ENTITIES

ARTICLE I
GENERAL PROVISIONS

Section 1. Purpose

These Rules provide the framework for the approval, execution, and deployment of Advanced Metering Infrastructure (AMI) provided by the Distribution Utilities (DUs) and ERC-Authorized Entities to promote energy efficiency, enhance system reliability and resilience, strengthen security measures, improve customer engagement, and advance the development of an intelligent and sustainable distribution network.

Section 2. Objectives

The objectives of these Rules are as follows:

1. To support the realization of customer benefits through improved electricity services, including accurate billing, strengthened customer service, and greater access to real-time energy data that empowers consumers to manage their energy consumption;

2. To promote demand-side management and energy efficiency by providing tools and services that increase customer awareness, visibility, and control over energy use;

3. To enhance the operational efficiency and network reliability of DUs by optimizing load management, mitigating energy theft– thereby reducing system losses, and shortening the power interruption duration;

4. To provide clear guidance to DUs and Entities for the effective implementation of AMI, including its associated features, service levels, and functionalities, in accordance with the applicable standards and regulatory requirements;

5. To establish technical standards for the components of an AMI project, including smart meters and related devices, communication platforms, and software applications necessary for its effective operation;

6. To strengthen consumer protection by ensuring strict compliance with existing rules and regulations on cybersecurity and data privacy relative to AMI deployment and operation;

7. To support the implementation of retail and market development programs such as Retail Competition and Open Access (RCOA), Retail Aggregation, Green Energy Option Program (GEOP), Net-Metering program for Renewable Energy, Smart Distribution Utility Roadmap (SDUR), enhanced Wholesale Market Design, and such other similar programs as may be adopted by the DOE and ERC;

8. To address operational challenges arising from disruptive events, including pandemics, natural disasters, and other crisis, through the use of AMI-enabled remote meter reading to ensure the accuracy and timeliness of billing and related services;

9. To promote efficient and sustainable electricity distribution through enhanced data collection and monitoring, improved grid management, revenue protection, renewable energy integration, and grid modernization, thereby ensuring a robust, reliable, and future-ready electricity system consistent with the regulatory objectives of affordability, reliability, and sustainability; and

10. To promote the deployment of smart and sustainable energy solutions in off-grid and underserved areas through the integration of Microgrid Systems, advanced technologies, and renewable energy systems, ensuring reliable, affordable, and inclusive electricity access at the least cost manner.

Section 3. Scope

These Rules shall apply to the following:

1. All DUs within their respective franchise areas, including Microgrid System Providers (MGSPs);

2. Retail Electricity Suppliers (RES) offering AMI supplemental services; and

3. All customer classes, except as otherwise provided for in these Rules and other existing ERC rules and regulations.

Section 4. Definition of Terms

Advanced Metering Infrastructure (AMI) refers to an integrated system that is typically comprised of smart meters and other related devices, communication platforms and/or a combination thereof, and meter data collection and management systems. AMI is a fundamental component of a smart network in which metering data is transported via wired or wireless means at a defined interval from the customer smart meter to the system.

Advanced Distribution Management System (ADMS) refers to a platform that integrates hardware and multiple utility-based systems which provides automated outage restoration and optimization of distribution grid performance.

American National Standards Institute (ANSI) refers to a private non-profit organization that oversees the creation, promulgation, and use of voluntary consensus standards for products, services, processes, systems, and personnel in the United States.

AMI-related Capital Expenditure (CAPEX) Plan/Projects refers to those projects for the implementation of AMI that are intended to serve existing and forecasted future loads, implement new solutions, and executing programs to maintain good and sufficient electric service to existing and future customers satisfying the technical criteria for capacity, reliability, quality, and safety.

Collector or Concentrator refers to a field functional equipment present in some AMI technologies that provides the communication interface between the smart meters and the Head End System. Collectors receive and collect the data from the meter and send it to the Head End System.

Communication Platform refers to a combination of network elements that enable two-way communication to and from the meter and the data center.

Consumer Meter Data Channel (CMDC) refers to a mechanism or means (e.g., SMS, Web Portal, Mobile App and Email) for customers to access and retrieve their meter data for their monitoring and to aid them in effectively managing their energy usage. The process can either be pushing the data directly to the CMDC, or for the customers pulling the data from the CMDC, or a combination of both.

Customer refers to any person, natural or juridical, who is the registered customer of the DU, or any person authorized by the registered customer to occupy the premises and enjoy electric service being supplied with electricity.

Cybersecurity refers to the application of technologies, processes and controls to protect systems, networks, programs, devices and data from cyberattacks.

Data Center refers to a centralized repository of servers, storage and Information and Communication Technology (ICT) network, either physical or virtual, responsible for data dissemination and information organized around a particular business.

Demand Control for purposes of these Rules, refers to the feature of the AMI that allows the DU to limit, interrupt and/or restore power to the identified load for a period of time.

Demand Response (DR) refers to a voluntary program where end-users change their usage pattern to manage electricity consumption in response to market prices or when system reliability is endangered. Demand Response includes all induced or intentional modifications to the consumption pattern of the customer to alter the timing, level of demand, or the total electricity consumption.

Demand Side Management refers to measures undertaken by distribution utilities to encourage end-users in the proper management of their load to achieve efficiency in the utilization of fixed infrastructures in the system.

Device Language Message Specification/Companion Specification (DLMS/COSEM) refers to a global specification that defines a language for data exchange between devices and/or Head-End Systems and guides manufacturers, vendors, and customers to ensure their devices and services meet certain criteria for data exchange, interoperability, efficiency, and

security.[1]

Direct Load Control refers to a demand response activity by which, as agreed upon by both parties, the customer's electrical equipment is remotely shut down or cycled on short notice.

Distributed Energy Resources (DER) refers to power sources connected to the distribution system or electrical system of the end-users that could be aggregated to meet a demand.

Distributed Energy Resource Management System (DERMS) refers to a software-based platform that controls and optimizes a group of DER assets including rooftop solar PV, behind-the-meter batteries, or a fleet of electric vehicles to balance power supply and demand and to deliver grid-based services.

Distribution Management System (DMS) refers to a collection of software applications designed to monitor and control the entire distribution network efficiently and reliably.

Distribution Utility (DU) refers to any electric cooperative, private corporation, government-owned utility or existing local government unit which has an exclusive franchise to operate a distribution system in accordance with its franchise and RA 9136. For purposes of this Rules, DUs shall include Microgrid System Providers (MGSPs).

DSOAR refers to the Distribution Services and Open Access Rules, including any amendments thereto.

Dynamic Operating Envelope System (DOES) refers to the system that calculates the dynamic limit of each customer's point of connection to the grid that can change in response to network condition, geographic location and time of day.

Elevated Metering Center (EMC) refers to the structure attached to DU's poles or other structures on which a cluster of meters is installed beyond the mounting height prescribed under the Magna Carta for Residential Electricity Consumers and Amended DSOAR provisions.

[1] DLMS UA (2023) - https://www.dlms.com/#section_ee76f251.

Enterprise Service Bus (ESB) refers to an architectural pattern implemented through middleware that enables integration between centralized software components and applications.

European Union Agency for Network and Information Security (ENISA) refers to a center of network and information security expertise for the EU, its member states, the private sector and Europe's citizens. ENISA works with these groups to develop advice and recommendations on good practice in information security. It assists EU member states in implementing relevant EU legislation and works to improve the resilience of Europe's critical information infrastructure and networks. ENISA seeks to enhance existing expertise in EU member states by supporting the development of cross-border communities committed to improving network and information security throughout the EU.

Export Energy refers to energy exported or delivered by the customer to the Distribution System.

Fault Location, Isolation, and Service Restoration (FLISR) refers to various technologies and systems which work in tandem to locate a fault, reduce both the impact and length of power interruptions by isolating the affected area(s), and automate power restoration.

Fiber Gigabit Passive Optical Network (GPON) refers to a field area communication technology that utilizes fiber optic cables for two-way data exchange between multiple sources and a receiver.

Field Area Network (FAN) refers to a network that provides the communications backhaul link from the collectors or concentrators to the Wide Area Network.

Geographic Information System (GIS) refers to a collection of computer hardware, software and geographic data for capturing, managing, analyzing and displaying every form of geographically referenced information, often called spatial data.

Head-End System (HES) or Meter Data Collection System (MDCS) refers to the system responsible for collecting meter data automatically. It also monitors and manages endpoint devices such as smart meters, communication modules and gateways.

Home Area Network (HAN) refers to a voluntary program to provide customers with the ability to manage and control their connected loads through a system provided by the DU or RES.

Import Energy refers to energy imported or received by the customer from the Distribution System.

Integrated Volt/Var Optimization System (IVVO) refers to the system that optimizes reactive power and voltage regulation resources to correct power factor and/or voltage issues and to reduce peak demand, energy losses and energy consumption.

International Electrotechnical Commission (IEC) refers to a worldwide organization for International Standards and Conformity Assessment for all electrical, electronic and related technologies also known collectively as electrotechnology.

Interruptible Load refers to a demand response program in which electric consumption of the customers is subjected to curtailment or interruption by the DU under tariffs or contracts that provide a rate discount or bill credit for agreeing to reduce load during system contingencies.

Interval Reading refers to any metered data (kWh consumption data, or kW demand) collected and recorded at fixed time intervals.

Magna Carta refers to the rules and regulations entitled *"Magna Carta for Residential Electricity Consumers,"* the guidelines implementing the same, and any amendments thereto issued by the ERC. It contains the rights and obligations of residential electricity consumers, ensuring access to fair, reliable, and transparent service while upholding due process and promoting accountability among Distribution Utilities (DUs).

Meter Communication Module refers to the network interface card for smart meters which enables the smart meter to communicate with other components of the AMI system using standard communications technologies, such as but not limited to Cellular, Fiber, Radio Frequency (RF) Mesh, PLC, NB-IoT, and LoRaWAN.

Meter Data Analytics System (MDAS) refers to a system or functionality/component of a system used in the analysis of metering data and its conversion to actionable information such as but not limited to,

energy consumption behavior, tamper detection, outage management, automated demand response, or energy feedback.

Meter Data Management Plan refers to planning documentation that covers the creation, protection, security, access, and management of consumer data generated by AMI systems.

Meter Data Management System (MDMS) refers to the system that aggregates, validates, estimates and edits (VEE) meter data such as energy export and import, and meter logs. An MDMS stores this data for a limited amount of time before it goes to a data warehouse and makes this data available to authorized systems.

Meter Installation Inquiry refers to retrieving meter data on demand in order to obtain the real-time key data needed for effective customer service (meter status, event log, etc.), outage management (confirmation all meters are restored) and operation (solar PV generation detection during load shedding), and other beneficial applications.

Metering Service Provider (MSP) refers to a person or entity authorized by the ERC to provide metering services.

Microgrid System refers to a group of interconnected loads and a generation facility with clearly defined electrical boundaries that acts as an integrated power generation and distribution system, whether or not connected to a distribution or transmission system.

Microgrid System Provider (MGSP) refers to a natural or juridical person whose business includes the installation, operation, and maintenance of Microgrid Systems in unserved or underserved areas nationwide.[2]

National Institute of Standards and Technology Interagency Report 7268 (NISTIR 7268) refers to the Smart Grid Cybersecurity guidelines covering security requirements, guidance on technical, process, organizational areas.

Neighborhood Area Network (NAN) refers to the communications leg that provides the link between the smart meters and field communication network gateways or devices.

[2] Definition adapted from Republic Act No. 11646, Microgrid Systems Act (2022).

Net Metering refers to a system appropriate for a distributed generation, in which a distribution grid user has a two-way connection to the grid and is only charged or credited the difference between its import energy and export energy.

Off-Cycle Reading refers to the remote reading of meters anytime within a billing cycle.

Operation and Maintenance (O&M) refers to the functions, duties and labor associated with the daily operations and normal repairs, replacement of parts and structural components, and other activities needed to preserve an asset so that it continues to provide acceptable services and achieves its expected life.

Outage Detection refers to the feature of a smart meter through the AMI which gives information to the DU of any customer experiencing power outage.

Outage Management refers to a system that can be used to optimize, automate, and efficiently manage outage and restoration of the distribution grid using data from field sensors and smart meter data.

Outage Management System (OMS) refers to a utility network management software application that models network topology for safe and efficient field operations related to outage restoration. OMS is integrated with contact centers to provide timely, accurate, customer-specific outage information, as well as with the Supervisory Control and Data Acquisition (SCADA) systems for real-time confirmed switching with breaker operations. These systems track, group, and display outages to safely and efficiently manage service restoration activities.

Philippine Distribution Code (PDC) refers to the set of rules, requirements, procedures, and standards, and any amendments thereto, governing Distribution Utilities and Users in the operation, maintenance, and development of their Distribution Systems. It also defines and establishes the relationship of the Distribution Systems with the facilities or installations of the parties connected thereto.

Platform as a Service (PaaS) refers to a cloud computing model that provides customers a complete cloud platform (for hardware, software, database, and infrastructure) for developing, running, and managing applications.

Power Line Carrier (PLC) refers to a communication technology that enables the sending of data over electric power lines, while simultaneously delivering electric power.

Prepaid Retail Electric Service (PRES) refers to an electric service using a Prepaid Metering System (PMS) designed to allow a customer to purchase credit and then use electricity until such time as the credit is exhausted.

Radio Frequency (RF) Mesh refers to a communication technology that interconnects each node in a mesh topology through wireless means such as radio applications of microwave, narrowband, and broadband among others.

Remote Configuration refers to the ability to adjust specific operational parameters of a smart meter's settings via remote capability while preserving the accuracy of meter readings, energy data, and existing recorded information. It includes logging of the date, time, and details of the changes made for transparency and accountability purposes.

Remote Connection refers to the connection of a new customer, following the physical installation of the metering facilities, to the distribution system using the automated features of a smart meter and without the need to physically access the customer's premises.

Remote Disconnection refers to the interruption/disconnection of power to the customer's property via the AMI meter's remote service disconnect switch through the AMI system without the need to physically access the customer's meter and/or disconnecting means.

Remote Reading refers to meter reading and recording using the automated features of a smart meter and without the need to physically access the customer's premises.

Remote Reconnection refers to the restoration of power to the customer's property via the AMI meter's remote service disconnect switch through the AMI system without the need to physically access the customer's meter and/or disconnecting means.

Retail Electricity Supplier (RES) refers to any person or entity authorized by the ERC to sell, broker, market or aggregate electricity to the end-users.

Service-Oriented Architecture refers to a way to integrate distributed, separately maintained and deployed software components, so that they can be rapidly incorporated into new applications without having to perform deep integration each time.

Smart Grid refers to a modernized electric grid that utilizes innovative technologies including two-way or multi-way communication, real-time monitoring, and advanced control systems, to achieve the following:

a. Improve overall reliability, power quality, security, efficiency and grid management ensuring full cybersecurity and interoperability;

b. Enable the seamless integration of bulk and flexible generation, Distributed Energy Resources (DER), Microgrid System, and electric vehicles through advanced management systems;

c. Empower customers by providing timely information and control options through enhanced energy management systems; and

d. Contribute to environmental sustainability by reducing the grid's carbon footprint through efficient energy usage, increased renewable energy integration, and optimized system operations.

Smart Grid Roadmap refers to the policy framework and phased implementation plan issued under DOE Department Circular No. DC2020-02-0003, which establishes the guiding principles, infrastructure components, and milestones for the transition of DUs and other stakeholders toward a smart and modernized electric power distribution system by 2040.

Smart Meter refers to an advanced technology electric meter capable of two-way communication that can measure, record, and transmit time-varying energy usage data and its components, derivatives, and events/logs. It includes a communications module for remote access function and may also serve as a gateway between the utility, customer site, and customer's Home Area Networking devices and/or load controllers.

Software as a Service (SaaS) refers to a full application that is managed and hosted by the provider. Customers access it with a web browser, mobile app, or a lightweight client application.

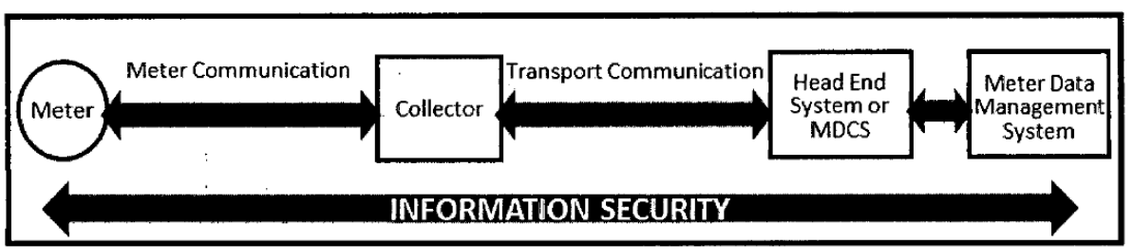
Time of Use (TOU) refers to a service that allows DUs to offer energy pricing scheme(s) based on the time of day when electricity is generated and the cost of supplying the same to customers during these specific periods, as agreed upon by both parties.

Wide Area Network (WAN) refers to the transport backbone network that connects all facilities to the backend systems including data centers.

ARTICLE II
BUILDING BLOCKS OF AN AMI

Section 1. AMI End-to-End Architecture

As defined in these Rules, the AMI architecture shall consist of Smart Meter(s), Collector or Concentrator (if applicable), Head-End System (HES) or Meter Data Collection System (MDCS), Meter Data Management System (MDMS), and the requisite Communication Platforms for meter data collection, transmission, and backhauling to the Data Center and/or Cloud Infrastructure. This may also include instances where third-party PaaS or SaaS configurations are utilized.



A Typical Architecture of an AMI

Relative to the foregoing, the AMI project shall include the Meter Data Analytics System, Smart Grid Roadmap Systems integration, and other components or systems necessary to implement the basic and supplementary services.

The DU's data center, Cloud setup and/or third-party PaaS or SaaS shall have a high-availability infrastructure or a disaster recovery site with recovery time objective and recovery point objectives that meet the minimum service level specifications provided in Appendix B.

The recovery time and recovery point objectives shall encompass the timeframe required to restore data center operations in the aftermath of a disaster.

Section 2. Smart Meters and Other Devices

A DU shall utilize smart meters and other devices that will fully operationalize the AMI.

A smart meter shall, at least, have the basic minimum technical specifications to support the delivery of the basic AMI services outlined in Article III, which include the following:

a. Active Energy and Demand "kWh-kW" meter;

b. Reactive energy and demand "kVARh-kVA" meter (where applicable);

c. Import and export measurement (where applicable);

d. Remote configuration and firmware update. The Remote configuration minimum functional specifications and service levels are set out in Appendices A and B, respectively;

e. Operating voltage in accordance with ANSI or IEC standards;

f. Frequency in accordance with ANSI or IEC standards;

g. Load Performance Accuracy in accordance with ERC Resolution No. 12, Series of 2009, as amended, or any amendments thereto; and

h. Disconnecting device, subject to the exceptions provided under Article III, Sections 1 and 2 of these Rules.

Other features, including but not limited to reactive energy and demand "kVARh-kVA" meter, RE System Generation Measurement, voltage quality detection, demand control, DERM, and compatibility with a customer's Home Area Network, may be optionally provided by the DU as part of the supplemental services.

Section 3. Collectors and Concentrators

Collectors or Concentrators may be installed for the purpose of collecting data from multiple smart meters and serve as the gateway between the meters and the Head-End System. Features may include the following: Alarm and Events (i.e., Power and Communication Outage/Restoration), Battery back-up.

Collectors and Concentrators are Field Area Network devices that may be backhauled using various wired and wireless packet-based solutions such as, but not limited to Wireless Private Network, fiber GPON, Public 3G/4G/5G, Ethernet, and future developments in communication technologies.

Collectors, Smart Meters – that receive and broadcast information over the air – and Concentrators shall require a type approval certificate from the NTC.

Section 4. Communication Platforms

An AMI project may utilize one or more existing communication platforms in its WAN, FAN, and NAN including, but not limited to, wireless (e.g., radio frequency and cellular solutions) or wired [e.g., fiber, Power Line Carriers (PLC), Ethernet, and RS 485] options, as well as new generation communication platforms developed through technological advancements after the effectivity of these Rules.

For communication platforms that use RF, the DU shall be responsible to secure approval from the NTC for the use of the frequency spectrum band.

Communication networks and equipment, depending on the communication platform utilized by the utility, shall be in place to enable two-way communication.

Synergy between smart grid, network control and field force communication facilities and requirements shall be taken into consideration to ensure the lowest overall cost to the customer.

Section 5. Head-End System (HES) or Meter Data Collection System (MDCS)

The DU shall install a HES or MDCS that will serve as the main user interface. The HES or MDCS shall be able to support the DU's AMI services which may include but are not limited to the following: remote upgrade of software/firmware; remote configuration of smart meters; collectors or concentrators; network and device management capability; scheduled, on-request and end-point initiated data collection and meter installation enquiry; remote service switch operation, and power outage/restoration reporting.

Other features, including, but not limited to DER Management, Remote Controlled Load Circuit Switch Operation may be made available by the DU or RES as part of the AMI supplemental services.

Section 6. Meter Data Management System (MDMS)

The DU or Metering Service Provider (MSP) shall be responsible for the installation of an MDMS to effectively manage the metering data collected by the AMI in accordance with these Rules, as well as the existing ERC rules and regulations such as PRES, RCOA, Retail Aggregation, and among others.

The MDMS must be seamlessly integrated with the systems of the DU, including their asset management system, customer information system, HES/MDCS, and Advanced Distribution Management System (ADMS). This integration facilitates the deployment of other supplemental AMI capabilities and services and ensures the overall cost-effectiveness of service provision to customers.

Moreover, the MDMS shall conform to data format standards, notably the ANSI data format standards detailed in Article II, Section 7. Alternatively, it shall align with other recognized data format standards such as the IEC 62056, IEC 61968, and other equivalent international standards.

Section 7. Interoperability

The AMI shall be capable of interoperating with different brands of smart meters and their corresponding HES to the MDMS. Other systems enabling AMI services offered by various vendors must likewise be able to

interoperate and integrate with the MDMS. In addition, a HES capable of handling multiple communication platforms and multiple meter brands simultaneously shall also be capable to interoperate with the MDMS.

To facilitate interoperability, data shall be structured to comply with the latest ANSI C12.19 standards, IEC 62056 or other equivalent international standard to allow for standardization of data tables and terms across AMI devices and systems, including integrated smart grid applications and other touchpoints.

All data communications shall also comply with the latest ANSI C12.22 standards, IEC 61970 or other equivalent international standard to ensure confidentiality, integrity, and interoperability of data over communications networks.

DUs may suggest alternate data format standards based on justifiable reasons. The implementation of such data formats shall be subject to the approval of the ERC.

Section 8. AMI System Security

DUs shall establish essential cybersecurity measures for the implementation of AMI. The primary aim is to safeguard AMI systems and data—ensuring their confidentiality, integrity, and availability—by referencing established standards and best practices. These Rules specify minimum cybersecurity requirements for critical AMI infrastructure components, acknowledging the evolving cyber threat landscape, diverse vendor offerings, and the imperative for practical implementation.

8.1. Required AMI Cybersecurity Program Elements

DUs shall adopt and implement a phased roadmap for their AMI cybersecurity program, designed to systematically develop mature cybersecurity resilience and ensure progressive alignment with relevant international standards. DUs shall structure their cybersecurity program to be consistent with local regulatory acts or circulars. The AMI cybersecurity program should integrate foundational organizational security measures with secure system operations and robust resilience capabilities and shall incorporate the following core elements:

a. Key foundational elements involve maintaining an inventory of AMI assets with secure configurations, implementing identity and access management (IAM) utilizing principles like least privilege and authentication, and ensuring comprehensive data security and privacy, particularly for customer Personally Identifiable Information (PII);

b. Secure system operations are achieved by embedding security throughout the system lifecycle, managing supply chain risks, and conducting proactive vulnerability management; and

c. The program must incorporate practical security monitoring, a documented incident response plan, foundational Business Continuity and Disaster Recovery (BCDR) strategies, and continuous security awareness training for personnel.

DUs shall undertake efforts towards securing global certification, such as ISO/IEC 27001 (Information Security Management System - ISMS) or equivalent industry standards, in parallel with or subsequent to the initial AMI project implementation. Unless specifically required under these Rules, such certification is not an upfront prerequisite for commencing an AMI project. To ensure the security, integrity, and resilience of the AMI system from its inception and until formal certification is achieved, the ERC shall require DUs to adhere to the provisions of Department of Energy (DOE) Circular No. DC2025-01-0001, entitled *"Institutionalizing the Energy Sector Cybersecurity and Cyber Resilience Framework"* and other future circulars or issuances of relevant regulatory agencies.

8.2. Minimum Cybersecurity Safeguards for AMI Infrastructure Components and Required Certifications

The AMI System shall provide stringent security measures throughout the network at all levels. Security features shall be integrated into meters, radio communication networks, collectors and HES. Data, while residing in the meter and during transmission to the Collector, must be protected. Additional security protection at the Head-End System(s) must likewise be put in place.

The AMI components should be provided with the following minimum cybersecurity safeguards:

a. Smart Meters: Secure boot mechanisms, firmware integrity validation, tamper detection, and secure remote management capabilities utilizing authenticated and encrypted communications, where such secure communication may be facilitated by an integrated component (e.g., a Network Interface Card);

b. Data Collectors/Concentrators: Secure configuration and hardening, physical protection, use of secure communication protocols, and data protection mechanisms;

c. Meter Data Management System (MDMS): Role-based access controls, protected audit trails, data integrity safeguards, and tested backup/disaster recovery procedures;

d. Head-End System (HES): Hardened operating environment, secure logging, robust credential management, appropriate network segmentation, design for high availability, and deploying intrusion detection/prevention capabilities, as applicable; and

e. Communications Systems: Utilization of network segmentation to isolate traffic and demonstrating secure implementation of communication protocols and data exchange, particularly for power system protocols such as DLMS/COSEM (IEC 62056) or ANSI C12 series.

In the event of any overlap, conflict, or inconsistency between the safeguards outlined herein and other applicable and equivalent standards, the requirements of relevant, globally recognized cybersecurity standards (such as those published by ISO/IEC 27001, or IEC 62443, or ANSI C12, as applicable to the specific component or system) shall take precedence to ensure the highest level of security and interoperability.

Compliance shall be required to any of the latest versions of the security standards, but not limited to the following:

a. ISO/IEC 27000 – Family of Standards and other relevant International Standards;

b. IEC 62443 – Global Standard for the Security of ICS Network;

c. ANSI C12;

d. IEC 62056 (DLMS/COSEM);

e. AMI-SEC System Security Requirements (America);

f. ENISA Smart Grid Security (Europe);

g. NIST Cybersecurity Framework;

h. NISTIR 7628 – Guidelines for Smart Grid Cybersecurity; and

i. Zero Trust Security Framework.

Proof of compliance to the foregoing standards shall be verified through third-party attestations and/or certifications meeting stringent integrity requirements. The DU must demonstrate adherence to any of the standards listed in the above paragraphs (a) to (d), as applicable or an equivalent internationally recognized cybersecurity certification. This demonstration can be achieved through third-party certifications or attestations, as applicable for the chosen standard. DUs should incorporate clauses into their vendor agreements covering key security assurances.

In summary, the minimum requirements should have addressed the following:

a. Warrant all hardware, software, and firmware are free from malicious code and do not contain undocumented functionalities, covert communication channels, unauthorized access mechanisms, or other characteristics

that could intentionally or unintentionally undermine the security or operational integrity of the AMI system;

b. Provide robust and verifiable methods to ensure the authenticity and integrity of all supplied hardware components, software, and firmware, including updates and patches, throughout their life cycle;

c. Demonstrate and maintain Secure Product Development Lifecycle (SPDL) and secure manufacturing practices consistent with recognized international standards, supporting "security and privacy by design" principles; and

d. Support DU's risk management efforts by maintaining transparency regarding the provenance of critical components and by cooperating with the DU to provide further assurance, which may include facilitating or sharing results from independent security assessments of critical components if reasonably required by the DU.

Section 9. Meter Data Analytics System (MDAS)

The DU's AMI-related CAPEX Plans/Projects shall incorporate provisions for future implementation of data analytics features and capabilities. These provisions are intended to facilitate the efficient analysis of data generated by the AMI system, enhancing various aspects of network planning, asset management, engineering, operations, and investments. Furthermore, they are expected to enhance customer service and improve the planning, design, and delivery of products, programs, and services.

The MDAS shall demonstrate the capability to seamlessly integrate with various systems, enabling the utilization of big data analytics.

Section 10. AMI Smart Grid Integration

A DU's AMI system shall integrate with other backend systems, especially smart grid systems, using a best practice integration system [e.g., a services-oriented architecture, featuring, and Enterprise Services Bus (ESB) and/or a real-time services bus for high priority communications].

ARTICLE III
BASIC AMI SERVICE AND CONDITIONS FOR DEPLOYMENT

An AMI shall have the ability to deliver the basic services as described in this Article.

Section 1. Remote Reading

An AMI shall allow the reading of a customer's smart meter remotely. The remote reading shall be conducted at the end of the customer's billing period and the data obtained therefrom can be used as the customer's monthly consumption subject for billing and payment.

At the time of approval of the application for electric service, the concerned customer shall be given the option to receive their monthly registered read data. If the customer elected to receive the same and the customer's Statement of Account (SOA) is not issued within forty-eight (48) hours from the conduct of the remote reading, the aforementioned registered read data shall be made available after the 48-hour period through the CMDC. The said 48-hour period shall be extended for justifiable reasons.

Notwithstanding the foregoing, the DU may conduct off-cycle remote readings.

In the event of any AMI-system malfunction which renders remote reading not possible, the DU may be allowed to conduct a manual reading of the smart meter.

Section 2. Remote Connection

An AMI must be capable of allowing remote connection of the customer's electric service.

Remote connection shall be undertaken upon the initial energization of the customer's premises following the DU's or RES' approval of the said customer's application for electric service and installation of the metering facilities.

The DU shall have the option to use smart meters without this functionality under any of the following circumstances:

a. Due to technical constraints, such as but not limited to customers with Current Transformer (CT)-rated metering; or

b. Other situations, subject to the approval of the ERC.

In the event of any AMI system malfunction which renders remote connection not possible, the DU may be allowed to conduct a manual connection of the smart meter.

Section 3. Remote Disconnection

An AMI must be capable of allowing the disconnection of a customer's electric service remotely on grounds allowed under existing laws and ERC rules and regulations, including but not limited to RA 7832, otherwise known as the Anti-Electricity Pilferage Law and its Implementing Rules and Regulations, MCREC and Amended DSOAR.

Prior to the remote disconnection of the said customer, the procedures set forth under existing laws, rules and regulations must have been complied with. Furthermore, the DU shall send a Remote Disconnection Warning (RDW) to the customer, through the CMDC, that the electric service will be remotely disconnected after the lapse of twenty-four (24) hours from the issuance of such warning, should the customer still unable to pay the bill or obligation.

Furthermore, DUs may utilize the disconnect capability of smart meters without prior notice in situations involving safety concerns or force majeure events that pose risks to the general public and/or the electric grid.

Notwithstanding the foregoing, the DU shall have the option to use smart meters without this functionality under any of the following circumstances allowed under Article III, Section 2 of these Rules.

In the event of any AMI-system malfunction which renders remote disconnection not possible, the DU may be allowed to conduct a manual disconnection of the smart meter.

Section 4. Remote Reconnection

As a general rule, the remote reconnection of an electric service previously disconnected shall occur within one (1) hour from the moment the customer's arrears are fully settled in the billing system of the DU, provided that there are no technical constraints or customer errors, such as payments for the wrong accounts. The one-hour period may be extended for justifiable reasons.

Notwithstanding the foregoing, the DU shall have the discretion to use smart meters without this reconnection functionality under any of the circumstances allowed under Article III, Section 2 of these Rules.

In cases where an AMI-system malfunction renders remote reconnection not possible, the DU may proceed with a manual reconnection of the smart meter.

Section 5. Outage Detection

An AMI system shall be capable of detecting power outages through the utilization of smart meters' outage detection functionality. Upon detecting a power outage at a customer's premises, the AMI meter shall promptly transmit an alert to the HES of the DU. Furthermore, the customer shall be promptly notified of the power outage and, whenever possible, provided with information regarding its cause.

Section 6. Meter Irregularity Detection (MID)

An AMI must be capable of detecting the irregularities, such as but not limited to:

a. Removal of the meter cover while the meter is still attached to its base or installation;

b. Removal of the meter from its base or installation; and/or

c. Cessation of meter operation, other than due to disconnection and termination.

Nothing shall preclude the DU from implementing an AMI capable of detecting additional forms of meter irregularities or bypass. The smart meters' MID capabilities must be disclosed at the time of application.

Section 7. Consumption Monitoring and Reporting Mechanism

Customers shall be provided with CMDC to be used in monitoring for each billing period, the following information:

a. The energy consumption in kWh;

b. The cumulative energy consumption in kWh for the applicable billing period; and

c. The total monthly consumption in kWh, which is determined by calculating the difference between the current reading for the present billing period and the last reading for the immediately preceding billing period.

Customers shall be entitled to free access to the abovementioned data. The concerned DU shall establish the procedures of ensuring accessibility and availability of such data or information as an integral part of its AMI project application.

At a minimum, customers shall be provided access to their consumption data at least once daily. The ERC may prescribe amendments to this minimum requirement through subsequent issuances, taking into account international best practices, system capabilities, cost implications, and the overall direction of the Philippine energy data framework. Costs reasonably and prudently incurred in complying with this requirement may be proposed for recovery, subject to ERC evaluation and approval. Any request by a customer for data services or platforms beyond the prescribed minimum shall be on a cost-to-pay basis, chargeable to the requesting customer.

The required information shall be made accessible to the customer through CMDC. The DU shall exert all efforts to notify and obtain from the customer the latter's choice of CMDC.

To further assist customers in managing their consumption, the DU, at its discretion, may provide additional information, including but not limited to:

a. On-demand transaction history covering a period of not more than twenty-four (24) months (e.g., payment, disconnection, reconnection); and

b. Consumption forecast and other analytics.

The CMDC shall likewise be used to send warnings when the customer's electric service is scheduled for remote disconnection. The customer shall receive such warning not later than twenty-four (24) hours prior to the said scheduled remote disconnection.

Section 8. Meter Installation Inquiry

The AMI system shall have the ability to retrieve meter data on demand to obtain the real-time relevant data needed for effective customer service (meter status, event log, etc.), outage management (confirmation that all meters are restored), as well as other beneficial applications.

Meter Installation Inquiry minimum functional specifications and service levels are set out in Appendices A and B, respectively.

ARTICLE IV
SUPPLEMENTAL AMI SERVICES

In addition to the basic services, supplemental AMI-enhanced services may be delivered by the concerned DU in accordance with these Rules and other existing rules and regulations. These supplemental AMI services and capabilities include, but are not limited to, those set out in this Article and new AMI features and capabilities developed through technological advancements after the effectivity of these Rules.

Section 1. Advanced Outage Management (AOM)

Any DU may be allowed to put up additional hardware and software systems that can integrate to implement an Advanced Outage Management

(AOM) program. The AOM must be able to assist the DU to efficiently manage the duration and frequency of outages by utilizing devices capable of communication and the identification of outage locations.

The AOM program must be able to perform the following activities:

a. Analyze the number of customers affected as an indication of the extent of the power outage(s);

b. Inform customers of the power outage status and estimated restoration time; and

c. Identify customers without power to the meter within the duration specified in Appendix B of these Rules.

The power outage information may be communicated to the customers of the DU through any or a combination of, but not limited to, the following channels:

a. Short Messaging System (SMS);

b. E-mail;

c. Mobile Applications;

d. Web Portal; or

e. Social Media.

Section 2. Power Quality Management

The AMI system may be utilized to monitor the Power Quality (PQ) condition of circuit or service area, including any or a combination of, but not limited to, the following parameters:

a. Voltage variations;

b. Voltage unbalance;

c. Power factor;

d. Harmonic distortion; and

e. Frequency

The AMI system shall be integrated with smart grid applications, such as Integrated Volt-Var Optimization (IVVO) and Dynamic Operating Envelope System (DOES), to optimize voltage profiles, manage reactive power, and maximize available network capacity for DER exports to the grid. These functions may be further supported through interoperability with other advanced systems, including Advanced Distribution Management Systems (ADMS) and Distributed Energy Resource Management Systems (DERMS).

Moreover, smart meters may be equipped with an automatic overload protection feature that disconnects or limits load once consumption exceeds safe operating limits or the customer's contracted capacity. This functionality is intended to safeguard electrical facilities, prevent fire hazards, and protect public safety.

Section 3. Demand Response

Demand Response (DR) refers to a service allowing a DU and/or customer to adjust the customer's electrical load usage in response to fluctuations in electricity prices or incentive-based compensation. This service also contributes to grid stability during specific periods. Typical DR programs that the DU or service provider that utilizes AMI metering data could offer, including but not limited to, Time-of-Use (TOU), Critical Peak Pricing, Interruptible Load Program (ILP), and Direct Load Control.

Section 4. Home Area Network

The DU may be allowed to provide a home energy management system or facilitate the involvement of other Home Area Network (HAN) service providers, enabling customers to monitor and control their consumption down to the connected loads.

The HAN service may provide functionalities including, but not limited to the following:

a. Energy usage per appliance in kWh and estimated Peso equivalent;

b. Remote switching (on/off) of appliances; and

c. Scheduling of appliance switching (on/off).

Section 5. Net Metering and Distributed Energy Resources (DER)

Net Metering refers to services rendered to an electricity customer under which energy generated by the customer from an eligible on-site renewable generating facility (not exceeding 100 kW) and delivered to the distribution system, may be used to offset the customer's purchased energy, in accordance with applicable ERC rules and regulations.

On the other hand, a DER refers to an eligible on-site renewable generating facility with a capacity exceeding 100 kW but not more than one megawatt (1 MW) for on-grid DUs, and capacity ranging from zero (0) kW to one megawatt (1 MW) for facilities under the Small Power Utilities Group (SPUG).

The AMI may be extended to enhance Net Metering (including those customers for their sole use), DER, and other similar services by providing the DU with visibility over existing and prospective DER installations. Such visibility is essential to maintaining distribution system reliability and security. The AMI may also support demand management and enable the regulation of DER output through integration with DERMS or other smart grid systems, as necessary to ensure network stability.

Section 6. Prepayment or Prepaid Retail Electric Service (PRES)

A DU interested in providing Prepayment services through the use of AMI shall comply with the ERC PRES Rules under Resolution No. 17, Series of 2012 entitled *"A Resolution Adopting the Amendments to the Rules for Prepaid Retail Electricity Service using a Prepaid Metering System"* or any amendments thereto.

Section 7. Smart Streetlights

Smart Streetlights (SSL) refers to a service that enables the DU to remotely monitor and control streetlight operations. A dashboard shall be made available to provide data on energy consumption, real-time on/off status, and facilitate control functions, including remote switching, dimming and scheduling of luminaires.

The SSL service may provide functionalities including, but not limited to, the following:

a. Remote Monitoring (inventory and status of each streetlight);

b. Advanced outage detection & notification; and

c. Light dimming & scheduling.

Section 8. Electric Vehicle Charging Demand Management

Electric Vehicle (EV) Charging Demand Management may utilize AMI to capture real-time and interval-based data on the electricity consumption of Electric Vehicle Charging Stations (EVCS). Such data shall enable DUs to monitor EV charging loads, manage demand on the distribution network, and support operational planning, including the implementation of customer solutions such as Time-of-Use (TOU) pricing.

The data obtained through AMI may be integrated with management platforms, including but not limited to EV Supply Equipment Management System (EVSEMS), to enhance operational monitoring, reporting, and load management.

The functionalities of AMI for EV charging demand management shall include, but not be limited to, the following:

a. Monitoring of EVCS energy consumption on a real-time and interval basis;

b. Visibility of charging load patterns and operational status;

c. Generation of reports and analytics to support customer insights, system operations, and EV infrastructure planning;

d. Facilitation of load management and demand response strategies; and

e. Enabling TOU pricing and consumption-based services for EV charging activities.

The integration of AMI with EV charging demand management shall support consumer access to reliable and cost-efficient EV charging services while enabling DUs to manage charging demand, maintain grid stability, and respond effectively to the increasing adoption of electric mobility.

ARTICLE V
DU ADOPTION AND IMPLEMENTATION OF AMI

Section 1. CAPEX Approval Process

A DU intending to adopt and implement an AMI Project, including modifications thereto shall file an application for approval of its CAPEX Projects necessary for such implementation in accordance with existing ERC guidelines (CAPEX guidelines or RRDWR, as may be applicable).

Section 2. Contents of CAPEX Application

The application shall include the following information and documents:

a. The basic services to be implemented through AMI;

b. The supplemental service(s), if any, to be implemented alongside with the basic AMI services;

c. Written description of the technical specifications of its AMI components and a certification that each of the following components meet the minimum requirements specified in these Rules, *to wit*:

   i. The type and features of the smart meter and other devices to be used, including their asset lives, and other information;

   ii. The type of communication platforms to be used, including component asset lives, and other information, as well as other alternative AMI communication platform that were considered and evaluated in determining the proposed system;

   iii. The type of software systems to be used, including, but not

limited to Head-End System (HES), Meter Data Management System (MDMS), analytics software, and enterprise service bus (ESB) or other equivalent systems;

iv. The Smart Grid Roadmap Systems to be integrated with, including but not limited to GIS, OMS, DMS, ADMS (including FLISR, IVVO), DERMS, MDAS, and provide detailed information how they will align to the SDUR;

v. The system capabilities and performance levels with regard to those set out in the retail metering standards and procedures;

vi. The associated systems to be used; and

vii. Provide the Server Infrastructure setup, be it on premise, in-house Cloud, or through a third-party Platform as a Service (PaaS) or Software as a Service (SaaS) model.

d. The type of measurements (e.g., active power) and interval settings (e.g., 5 minutes, 15 minutes) that the system is capable of recording following remote configuration;

e. Smart Meter type approval from the ERC;

f. Radio communications component type approval from National Telecommunications Commission (NTC);

g. Compliance to the provisions of Article II, Section 8 hereof;

h. The cost of the proposed CAPEX and the benefits to be derived by the DU, as well as their customers, from the implementation of the proposed AMI CAPEX;

i. The proposed project implementation schedule, including the target date of completion;

j. A meter data management plan on how metering data will be created, stored, secured and managed; and

k. A communication plan on how to inform and educate the customers about the AMI.

Section 3. Testing and Inspection of AMI Hardware and Software

The ERC may conduct testing and inspection to validate the AMI Basic Services and/or Supplemental Services, in accordance with Appendix C of these Rules.

ARTICLE VI
REQUIREMENTS FOR SMART METER

Section 1. AMI-eligible Smart Meters

Only those types of meters with prior approval from ERC are eligible for use in AMI revenue metering. The smart meter shall be certified as being IEC or ANSI standard compliant.

Section 2. Accuracy Requirements for the Smart Meter and Reliability of Other Devices Used in the AMI

Smart meters, collectors/concentrators (if applicable) and the head-end system must be accurate and reliable.

Before installation, the smart meters must be tested and sealed by the ERC. The tolerance limits provided for in existing ERC rules and regulations shall apply to new smart meters as well as those smart meters already in service.

The applicant DU shall safeguard the confidentiality, integrity and accessibility of data transmitted from the meter to the MDMS. Confidentiality ensures that certain information is solely accessible to authorized persons and entities; integrity is upheld to ensure the authenticity and accurate representation of source data without unauthorized manipulations; while accessibility mandates that authorized persons and entities can retrieve the data as needed.

Section 3. Location of Smart Meters

As a general rule, smart meters shall be installed in a clean, vibration-free location that ensures accessibility and visibility for readings and testing by the DU and the customer, unless stated otherwise in these Rules. Nevertheless, for smart meters equipped with alternative display capabilities, the provision of information through an authorized load monitoring device, a Customer Meter Data Channel (CMDC), or a similar mechanism shall be considered compliant with the visibility requirement for reading.

Smart meters may also be located in other areas based on justifiable reasons. The provisions on the location and relocation of meters in existing ERC rules and regulations including but not limited to Article 11 of the Magna Carta for Residential Electricity Consumers (MCREC), Article 2.11 of the Amended Distribution Services and Open Access Rules (DSOAR) and Article 2.2 of the Rules on Elevated Metering Centers, and any amendments thereto, shall apply.

Section 4. No Meter Deposit Requirements for Smart Meters

A smart meter shall be provided to the customer by the DU and no meter deposit shall be collected from customers, consistent with the provisions of the Magna Carta, Amended DSOAR and/or Article XI, Section 2 of these Rules.

ARTICLE VII

ELECTRICITY PILFERAGE AND OTHER METER IRREGULARITIES IN AN AMI PROJECT

Section 1. Illegal Use of Electricity; Tampering of AMI Infrastructure Devices or Components

In relation to these Rules, the following acts shall be considered as illegal use of electricity in accordance with the provisions of Section 2 (c) and (d) and Section 4 (iv) of RA 7832, or any amendments thereto:

1. Tampering with the smart meter;

2. Altering data with incorrect information while the said data is at

rest in the smart meter or during transmission to the collector or concentrator, or otherwise injecting forged metering data into the communication between the smart meters and the DU's or authorized service provider's facilities; or

3. Acts that would deliberately interfere, obstruct or disrupt the communication of AMI infrastructure devices and/or components.

Section 2. Discovery and Confirmation of Tampering of Smart Meters

Discovery of a possible tampered smart meter shall be made in the presence of the required witnesses and in accordance with the procedures set forth in Republic Act No. 7832 and its Implementing Rules and Regulations, as amended.

Data logs or recorded events obtained from the AMI system may be used as supporting evidence in identifying possible tampering or irregularities, subject to validation and confirmation by the ERC in accordance with existing rules and procedures.

Tampering of a smart meter shall be confirmed only by the ERC. No defective or tampered meter shall be placed in service or be allowed to remain in service without adjustment and correction. Adjusted and corrected meters shall be subject to testing by the ERC prior to their reinstallation to the premises of the customers.

Section 3. Applicability of Republic Act No. 10175, otherwise known as the Cybercrime Prevention Act of 2012

The violations under Article VII, Section 1 hereof, are however, without prejudice to the application of the provisions of Republic Act No. 10173, otherwise known as the Data Privacy Act of 2012 and Republic Act No. 10175, otherwise known as the Cybercrime Prevention Act of 2012. Thus, the offending party or parties, for the same act, may be held liable for violations of RA 7832, RA 10173 and RA 10175.

Section 4. Billing Adjustment and Differential Billings in the AMI

The determination of defective smart meters and other metering facilities, as well as other billing errors, and the computation of the

corresponding billing adjustments, shall be made in accordance with the provisions of the MCREC, Amended DSOAR and other relevant laws, rules and regulations.

Conversely, the discovery and determination of tampered smart meters and other metering facilities and the corresponding differential billings shall be made in accordance with the procedures, methodologies and recovery periods set forth in RA 7832 and its implementing rules and regulations.

Section 5. Disconnection of a Customer with an Installed Smart Meter

A DU, after due notice to the customer, shall be allowed to remotely disconnect a customer for illegal use of electricity and for failure to pay the required differential billing within the prescribed period under the Implementing Rules and Regulations of RA 7832.

Section 6. The MID Warning; Consequences

To alert the DU of a possible commission of illegal use of electricity, the AMI must be capable of warning the said DU or authorized service provider of irregular meter operation detected in any of the circumstances provided for under Article III, Section 6 of these Rules.

Upon receipt of the warning, the DU may inspect the meter and its metering installation within a reasonable period of time. Failure to inspect within a reasonable period of time despite receiving warning shall constitute negligence on the part of the DU, which may limit the recovery of the differential billing in cases of discovery of illegal use of electricity upon such inspection.

Provided however, that the DU shall not be considered as negligent if the failure to inspect within a reasonable period of time, is due to any of the following circumstances:

1. The customer or its representative is not present during the conduct of inspection;

2. Where the DU have encountered or received large volume of MID warnings making it physically impossible to conduct the inspection

within said period;

3. A fortuitous event where there is no other opportunity for the DU to conduct the inspection within a reasonable period of time; and

4. Other analogous circumstances.

Alternatively, the DU may also be required to provide proof that the illegal use of electricity started earlier than the said MID warning. In any case, the period of differential billing recovery shall, in no case, be less than one (1) year preceding the date of discovery of the illegal use following the occurrence of the MID warning, except when there is indubitable proof that the illegal use occurred in less than a year.

ARTICLE VIII

THE AMI ARCHITECTURE AND SERVICES UNDER THE REGIME OF RETAIL COMPETITION AND OPEN ACCESS (RCOA)

Section 1. AMI Services of an RES

The DU shall allow an RES to integrate its AMI supplemental services to the DU's AMI architecture through commercial arrangements, the type of which shall be at the option of the DU.

Section 2. Rights and Obligations of an RES

An RES undertaking AMI supplemental services shall be bound by the same rights and obligations as the DUs which were previously authorized by the ERC to provide the aforementioned services. They must likewise comply with all relevant provisions of these Rules depending on the type of service they are providing to their respective contestable customers.

Section 3. Commercial Arrangements between DUs and RES

To operationalize the approved AMI supplemental services offered by an RES, DUs shall allow such RES to utilize the requisite hardware and software facilities incorporated into the DU's AMI through duly executed commercial arrangements.

In the event that an RES integrates its own system(s) with the DU's AMI, the commercial arrangements between both parties shall incorporate appropriate cybersecurity and data privacy measures to protect the integrity and confidentiality of both systems. The RES shall be granted access only to the data necessary for the performance of its AMI supplemental services to its contestable customers, consistent with the provisions of the Data Privacy Act and applicable ERC rules.

The type of commercial arrangement between the DU and an RES, under this Article, shall be determined by both parties, subject to fair, reasonable, and non-discriminatory principles, and shall be consistent with the competitive framework established under the RCOA regime. In case of disagreement, the ERC shall resolve the same pursuant to Section 43(U) of Republic Act No. 9136.

To facilitate regulatory oversight, the DU shall submit to the ERC a summary report of each executed commercial arrangement within **thirty (30) calendar days** from its effectivity. The report shall include the nature and scope of the agreement, system integration scope, cost-sharing arrangements, and data access protocols, for ERC's monitoring and compliance purposes.

ARTICLE IX

THE AMI ARCHITECTURE AND SERVICES SUPPORTING THE SMART GRID

Section 1. Technical and Deployment Solution Alignment with the Smart Distribution Roadmap

The proposed AMI technical solution and deployment program must seamlessly be aligned with and support the DU's SDUR. Specifically, the technical design, including data management, technical services, and system integration of AMI with smart grid elements, must be explicitly outlined. Additionally, the timing of integration shall be synchronized with the DU's other smart grid work streams and systems.

To ensure coherence with the SDUR, the AMI deployment program shall be delineated in phases, commencing with planning and pre-deployment activities. Subsequent phases shall involve piloting the AMI technical and deployment solution, scaling up the solution, and ultimately achieving full deployment of the AMI.

The AMI solution architecture shall therefore consider the full smart grid scope, ensuring that proposed technical and deployment solutions are optimized across the AMI and smart grid programs. All data required by smart grid systems shall be factored into the AMI technical solution in the most efficient manner possible.

ARTICLE X
AMI DEPLOYMENT

Section 1. AMI Deployment Planning, Risk Management and Governance

A comprehensive deployment plan shall be developed, encompassing essential aspects such as program overview, cost-benefit analysis, management strategy, envisioned end state ('to-be' state), delivery scope and approach. This plan must explicitly delineate the capabilities, resources, and budget allocation, as well as identify key risks and propose a robust risk management strategy. It shall address the management of sites that are challenging to access or communicate with and incorporate provisions for customer education, and customer data management. Additionally, the deployment plan must outline the governance approach of the program, inclusive of arrangements for performance reporting.

A phased strategy for AMI deployment shall be detailed within the plan, aligning program costs with anticipated benefits. This may involve the initial deployment of a cost-effective basic AMI meter, along with corresponding communications and IT solutions to support the basic services. Subsequent stages may include upgrading to more advanced AMI meters upon request by the customer of the DU, with associated costs transparently charged to the requesting party.

ARTICLE XI
MISCELLANEOUS PROVISIONS

Section 1. Use of Other Type-Approved Smart Meters

Nothing in these Rules shall preclude a DU from using any other type-approved meter in the implementation of its AMI project provided that the features of the new type of meter include the same functions as the meter-type specified in the AMI project application.

Section 2. Power Interruptions Due to System Maintenance or any AMI Malfunction

Information on scheduled power interruptions due to system maintenance must be sent to the customers in accordance with the prescribed period set forth under the MCREC and the Amended DSOAR.

In case a customer or group of customers experience unscheduled power interruptions resulting from a malfunction of the AMI hardware, software, or communication network, all efforts shall be exerted to restore power service within a reasonable period, consistent with the severity of the issue and any prevailing industry practice.

The ERC may, in future issuances or amendments to these Rules, prescribe specific restoration timeframes or performance standards based on operational experience and best international practices.

Section 3. Automated Meter Reading

Notwithstanding the provisions of these Rules and subject to the approval of the ERC, a DU may be allowed to implement a stand-alone Automated Meter Reading (AMR) project, utilizing a one-way communication to avail remote reading applications.

Section 4. Third-Party Partners, Contractors and/or Agents

A DU may be allowed to engage the services of third-parties, and/or hire contractors and agents in the implementation of their AMI project. However, the DU shall be primarily liable to the customer for any violation of the AMI project, as well as these Rules, committed by the said third-party partner(s), contractor(s) and/or agent(s).

Section 5. Customer Data and Information; Meter Data

All customer data and information collected by the DU, including its employees, contractors, agents and third-parties mentioned under Article XI, Section 4 hereof, in the course of implementing an AMI project, shall be treated as confidential. This confidentiality is in accordance with the provisions outlined in the Amended DSOAR and the Republic Act No. 10173, otherwise known as the Data Privacy Act.

Data obtained from the smart meter must be available within a period of at least five (5) years, provided, however that data from smart meters that are subject of complaints must be retained until the final resolution of the case. The said data and/or information may be stored in the MDMS or any archive of the concerned DU/Authorized Entities, with the confidentiality and integrity thereof protected at all times.

ARTICLE XII
FILING, PROCESSING AND FEES

Section 1. Filing and Processing of Applications

The filing and processing of applications for AMI-related CAPEX projects shall be governed by the *Amended Rules of Practice and Procedure of the ERC*, promulgated under Resolution No. 01, Series of 2021, or any amendments thereto.

Section 2. Application Fees

An application fee shall be imposed on all AMI-related CAPEX applications in accordance with Resolution No. 21, Series of 2007, entitled *"A Resolution Approving the Revised Schedule of ERC Fees & Charges,"* or any amendments thereto.

ARTICLE XIII
REPORTORIAL REQUIREMENTS AND FINES AND PENALTIES

Section 1. DU Reports

The concerned DUs implementing an AMI system shall submit to the ERC periodic reports containing the following information, except for retrofit programs which shall adhere to the minimum reporting requirements outlined in Appendix D, *to wit*:

1. Status of AMI implementation, including areas of deployment within the franchise area;

2. Number of customers per classification with installed smart meters

within the specific quarter and the accumulated total numbers from the time of AMI project implementation;

3. Number of sites with installed but non-operational smart meters;

4. Complaints and issues received in connection with the AMI implementation and actions taken, if any; and

5. An assessment of how AMI technology has improved overall network performance.

The AMI implementation report shall be submitted semi-annually for the first two (2) years from the date of project commencement–not later than January 31st and July 31st of each pertinent year–and annually thereafter, not later than March 31st of each pertinent year, or as may be prescribed by the ERC.

DUs shall likewise maintain records of meter events, including MID warnings that occurred in the last five (5) years. Such records shall be made available for periodic inspection or audit by the ERC.

Section 2. AMI Malfunction(s); Violation(s) of the AMI Rules and Approved AMI Project and Services

Any AMI system malfunction shall be resolved by the concerned DU within the indicative timeframe specified in its ERC-approved AMI application. Failure to resolve such malfunction within the prescribed period, without justifiable cause, shall be a ground for the imposition of appropriate fines and penalties in accordance with ERC Resolution No. 3, Series of 2009 entitled *"Resolution Amending the Guidelines to Govern the Imposition of Administrative Sanctions in the Form of Fines and Penalties pursuant to Section 46 of the Republic Act 9136,"* the Electric Power Industry Reform Act (EPIRA), and other applicable rules and regulations.

When the malfunction arises from causes beyond the DU's control, the same shall be promptly reported to the Commission, together with supporting documentation identifying the cause, corrective actions taken, and proposed preventive measures.

Any other violation of these Rules shall likewise be subject to administrative sanctions under the same regulatory framework.

ARTICLE XIV
FINAL PROVISIONS

Section 1. Capital and Operational Expenditures

The expenses associated with the procurement, installation, and commissioning of smart meters, devices, hardware, software and processes necessary for the implementation of an approved AMI project, whether offering basic and/or supplemental services, shall be recovered in accordance with the Rationalized Rules for Distribution Wheeling Rates (RRDWR) or RSEC-WR, as applicable, and other relevant ERC rules and regulations on capital and operational expenditure approval.

Section 2. Compliance with Other ERC Rules and Regulations

Approval under these Rules is without prejudice to the compliance of the applicant and authorized service provider with existing laws, ERC rules and regulations, including but not limited to the RRDWR or RSEC-WR, and other applicable rules governing capital and operational expenditure approvals, RA 7832 or otherwise known as the Anti-Electricity Pilferage Law, Amended DSOAR, MCREC, and RCOA.

Section 3. Transitory provision

All pending applications with the ERC for approval of an AMI project shall be dismissed without prejudice.

Section 4. Exception Clause

For good causes, including changes in government policies, the ERC may allow an exception from any provision of these Rules, if such exception is found to be in the public interest and is not contrary to law or any other pertinent rules and regulations.

Section 5. Separability Clause

If for any reason, any part or section of these Rules is declared unconstitutional or invalid, the parts or sections thereof which are not affected thereby shall continue to be in full force and effect, unless such declaration would render the entire Rules unenforceable or non-

implementable.

Section 6. Repealing Clause

The existing AMI Rules, other ERC rules, regulations, policies, decisions, guidelines or portions thereof, not consistent with these Rules, are hereby repealed or modified accordingly. Rules and regulations as well as provisions thereof not affected shall remain in force and effect. In the event that these Rules are suspended or rendered invalid by the Courts, the existing AMI Rules shall be reinstated and continue to be in full force and effect.

Section 7. Effectivity

These Rules shall take effect fifteen (15) days following its publication in a newspaper of general circulation or in the Official Gazette.

Pasig City.

MKCM / MCS / RFA / GDB/MCCG

Appendix A: Minimum Functional Specifications

Meter Installation Inquiry Minimum Functional Requirements		
1.0	Meter Installation Status	
	1.1	Serial Number
	1.2	Make
	1.3	Model
	1.4	Type
	1.5	Contactor Status
	1.6	Meter Status
2.0	Meter Readings	
	2.1	Registers
	2.2	kWh
	2.3	kVarh (if available)
	2.4	Average Voltage

Appendix B: Minimum Service Levels

Service	Completion Time	Completion Rate
Meter Read	10 minutes from the issuance of a command to the time the system acknowledges the execution	90% to 95% completed within the minimum service levels
Meter Installation Inquiry	10 minutes from the issuance of a command to the time the system acknowledges the execution	90% to 95% completed within the minimum service levels
Meter Disconnection	10 minutes from the issuance of a command to the time the system acknowledges the execution	90% to 95% completed within the minimum service levels
Meter Reconnection	10 minutes from the issuance of a command to the time the system acknowledges the execution	90% to 95% completed within the minimum service levels

Disaster Recovery Requirement	Minimum Service Level
Recovery Time Objective	4 hours
Recovery Point Objective	15 minutes

Appendix C: Minimum AMI Testing Requirements

Minimum AMI Testing Requirements		
1.0	Validation of Main Technical Assumptions and Risks	
	1.1	Validation of All Metering Configurations
	1.2	Validation of All Communications Pathways
	1.3	Validation of All AMI Services, including Service Levels
	1.4	Validation of CMDC
	1.5	Validation of AMI-based Billing Process
	1.6	Validation of Supported Smart Grid Application Functions
	1.7	Validation of End-to-End Supplementary Service Functions
	1.8	Validation of Security and Access Compliance
2.0	Validation of Main Economic Assumptions and Risks	
	2.1	Validation of Avoided Manual Billing Costs
	2.2	Validation of Installation Costs
	2.3	Validation of Fault Rates for Meters and Communication Devices
	2.4	Validation of SAIFI, SAIDI & EENS improvement (if applicable)
	2.5	Validation of System Loss Reduction (if applicable)
	2.6	Validation of Customer Energy Efficiency Assumptions (if applicable)
	2.7	Validation of Demand Response Assumptions (if applicable)

Appendix D: Minimum Retrofit Program Reporting

Minimum Retrofit Reporting Monthly	Month	
	1	...n
Program X...n		
Premises with Access Issues		
Premises with Communication Issues		
Premises with Meter Faults		
Premises with Customer Complaints		
Premises with 1-Phase Meter Installs		
Residential		
Commercial		
Industrial		
Premises with 3-Phase Meter Installs		
Residential		
Commercial		
Industrial		